Stable Answers, Unfinished Reasoning: Why Self-Consensus Is Not a Safe Early-Exit Signal

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Abstract

A natural way to cut reasoning-model inference cost is to repeatedly probe a single partial trajectory for its current answer and stop once probes *agree*—*self-consensus*. We ask whether any such rule is both safe and token-saving, and whether one can be selected once and reused. A *preregistered* sweep of 3,520 consensus rules, replayed on frozen trajectories from two models and three benchmarks, clears none of three acceptance gates fixed in advance; the frontier reproduces on a held-out split and on two unseen models—while a boundary-confidence control (DEER) swept through the same pipeline clears all three. The reason is that the signal measures the wrong object: agreement establishes that the current answer *persists* under a fixed probing procedure, not that the reasoning has *terminated*—a *consensus–termination gap*. Stopping on it destroys a correct-in-the-end answer $\mathbf{2\text{–}45}\times$ more often than it banks a wrong one, never approaching the 1:1 of sampling noise at any window that still saves tokens; probe re-wording and a hand-labelled error taxonomy show the agreed answer is often a placeholder the model had not settled on. Agreement fails not because it is insufficiently strict, but because it repeatedly measures the wrong object.

1 Introduction

Reasoning language models (DeepSeek-AI, 2025; Qwen Team, 2025; OpenAI, 2024) can spend thousands of tokens on a single answer. *Early exit*—halting a chain of thought once its reasoning is done—could cut that cost (Chen et al., 2024; Fu et al., 2024), but it requires knowing *when the reasoning has finished*, which the trajectory does not announce. A natural and increasingly studied proxy is *self-consensus*: repeatedly probing a *single* partial trajectory for its current answer (e.g., by appending a short "`Final Answer:`" suffix) and stopping once recent probe answers *agree* (Fu et al., 2024). We abbreviate self-consensus as *consensus* below, and reserve *self-consistency* for the distinct setting in which several trajectories are sampled independently (§2). The premise is that if the model repeatedly says x, it has probably finished reasoning about x; recent work reports answers that "converge" well before a trajectory ends and stops on that convergence at little *aggregate* accuracy cost (Liu and Wang, 2025).

We show that this proxy measures the wrong object. Under a fixed probing procedure, repeated agreement establishes only that the current answer *persists*, not that the reasoning has *terminated*, and the two come apart—a *consensus–termination gap*. This reconciles the aggregate savings above: those savings are real, but they ride on a *directional* accuracy cost. Aggregate accuracy alone can obscure this directional cost, while excluding probe overhead can overstate the resulting token savings.

The gap has a concrete source. A probe requires an answer even when the underlying trajectory has not settled, so repetition under a fixed query may reflect the elicitation procedure rather than reasoning termination; this elicited placeholder can repeat across many probes and look settled while the underlying trajectory goes on to correct it. Three direct measurements bear this out (Section 4; the first two are single-environment observations, the third spans all 18): re-probing the *same* frozen prefix with a differently worded query returns a different answer 54% of the time in the first tenth of a trajectory against 10% near the end, so early "answers" are largely artifacts of being asked; hand-labelling 134 wrong stops finds three in five commit to an answer the model had not converged on, against one in five it had genuinely settled; and consensus destroys a correct-in-the-end recovery $\mathbf{2\text{–}45}\times$ as often as it banks an answer from an ultimately incorrect trajectory, depending on how much agreement the rule demands. This is opposite to the aim of safe early exit: the removed computation is predominantly computation that would have corrected the answer. Widening the agreement window does not close the gap—it lowers the loss by firing less often and only later, so the $\sim 2\times$ end of that range buys a net saving of 7.9%, whereas DEER, the non-consensus control below, holds 2.4–3.5$\times$ while saving 28–32%.

Prior reports establish good accuracy–saving points for a particular model, benchmark and parameter setting. We ask two stricter questions: whether *any* rule in the family is simultaneously safe and saving at all, and whether one can be *selected* once and reused rather than retuned per model and benchmark. To answer both, and to rule out a badly chosen threshold or a single implementation, we search the signal exhaustively under commitment. On frozen trajectories from two models and three benchmarks we run a *preregistered* sweep of the windowed consensus family we study—window size $\times$ share threshold plus operational knobs, 3,520 rules—with problem-grouped splits and acceptance gates fixed in advance (Section 5). No setting of these knobs clears any gate: **not one of the 3,520 rules is both safe and saving**—not even on the split the gates are applied to, so nothing is selected to carry forward; of the 478 that clear a gate *in-sample* on train, none clears it on dev, the split selection runs on. The frontier reproduces on the test split (r=0.98) and on unseen models at $4\times$ the scale and a different architecture, and a shipped operating point (CertaIndex (CoT)) loses 56–70 pp on the same trajectories.

That the gap cannot be tuned away leaves one alternative explanation—that early exit is simply impossible at these budgets. It is not: swept through the same pipeline and gates, a non-consensus method—DEER, which reads the model's confidence at a reasoning boundary—clears all three, losing as little as 0.3 pp while saving 28%, on the unseen models too (§5.3). Figure 1 is the argument in one picture.

Contribution.

- **We identify a consensus–termination gap in probe-based early exit**: under a fixed probing procedure repeated agreement establishes only that the current answer persists, not that the reasoning has terminated (Section 4).
- **We explain what agreement actually measures**, through three direct signals—the wording sensitivity of the extracted answer, a hand-labelled error taxonomy of stopped-but-wrong cases (κ=0.82), and a trajectory-level recovery asymmetry—which together show that early agreement is a probe-elicited placeholder on an unfinished trajectory rather than a settled belief (Section 4).
- **We show the gap is not a threshold or implementation artifact**: a preregistered sweep of 3,520 rules on frozen trajectories closes no gate, while a non-consensus signal swept under the same protocol clears the same gates (Section 5). We release the protocol, trajectories, and sweep in full.

2 Related Work

Probe-based / consensus early exit. The methods we scrutinize periodically query a partial reasoning trajectory for its current answer and halt on agreement. Dynasor / CertaIndex (Fu et al., 2024) formalize this with a "certainty index" over recent probe answers and use it to terminate early when serving reasoning programs, a response to the *overthinking* of reasoning models (Chen et al., 2024; Yang et al., 2025). Recent work pushes the same idea further: Liu and Wang (2025) report that a trajectory's answer "converges" well before it ends and stop on that convergence at little *aggregate* accuracy cost, while learned prefix-feature stoppers reach the more cautious conclusion that no single aggressive policy is universally safe (Dong et al., 2026). We adopt the same probe mechanism (a short boxed-answer suffix) and notion of self-consensus, and ask the question these leave open: whether *any* rule in this family is both safe and saving, or whether the apparent aggregate savings hide a directional accuracy cost. These results show that good accuracy–saving points exist for a given model, benchmark and parameter setting; we ask instead whether one *transferable policy* can be chosen on development environments and carried to new benchmarks, models and seeds without retuning.

Where agreement gets its meaning. Self-consistency (Wang et al., 2023) samples a *diverse* set of reasoning paths and takes the majority answer; agreement is evidence there because the paths are drawn independently, so the vote can outweigh an error on any single path, and early-stopping variants only decide how many samples to draw (Aggarwal et al., 2023; Li et al., 2024). Probing a

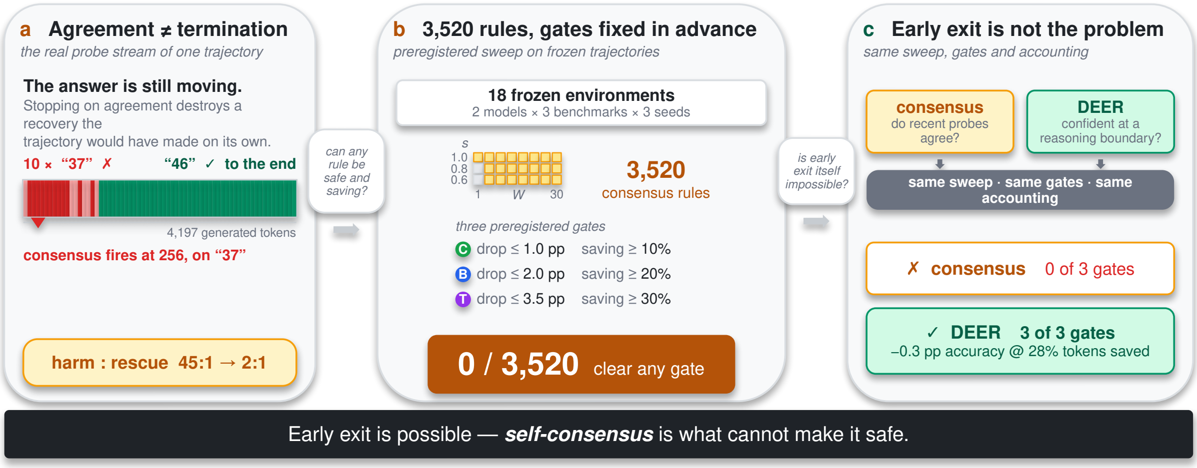

Figure 1: **The argument in one picture. (a)** On MATH500 problem 68 (DeepSeek-R1-Distill-Qwen-7B, dense 64-token probes) the model emits the same wrong answer, 37, for ten consecutive probes; a three-probe rule fires at token 256 and commits it. Left to run, the trajectory reaches the correct 46. Ribbon colours: red the dominant wrong answer, pink other wrong answers, green correct. **(b)** The preregistered sweep of 3,520 rules, replayed on 18 frozen environments against three gates fixed in advance. None clears any gate. **(c)** Swept through the same pipeline, gates and accounting, a boundary-confidence method (DEER) clears all three—on the test split, at 4× scale and on a different architecture too (§5.5).

single chain of thought—*self-consensus* (§1)—is a different object: successive probes observe nested prefixes from the same evolving reasoning path and therefore share most of their reasoning history, so agreement records that one line of reasoning has not changed its answer, not that separate lines of reasoning have converged—the protecting vote is absent. On its own, though, this structural point does not settle whether agreement tracks *termination*.

Alternative termination signals. A stop can read signals other than consensus: verifier- and process-reward-guided decoding (Lightman et al., 2024; Wang et al., 2024) scores partial reasoning, adaptive-computation methods learn a halting signal at the token or layer level (Graves, 2016; Schuster et al., 2022), confidence-dynamics stops read how the model's answer confidence evolves (Hosseini et al., 2026), and DEER (Yang et al., 2025) stops on the model's confidence in a trial answer at reasoning boundaries. We sweep DEER through our own pipeline in Section 5.3. We do not claim this broader class is categorically superior; our experiments isolate one signal.

3 Experimental Setup

Every measurement in this paper—the direct measurements of the gap (Section 4) and the rule sweep that tries to tune it away (Section 5)—runs on the same frozen substrate under the same token accounting, so that no result can be an artifact of a rule changing the reasoning it observes.

3.1 Frozen trajectories, offline probes

For each problem we generate exactly *one* main trajectory in a single request (caps: 16K tokens for MATH500 and AMC23, 32K for AIME24) and freeze it. All probing is then performed *offline* against fixed text prefixes of that frozen trajectory. Because probes never re-enter the decode, changing a probe schedule—or a stopping rule—cannot change the underlying reasoning; every rule is scored against the same trajectories. We build two offline probe banks: a *dense* bank that re-probes every 64 tokens with simple@32 (a 32-sample boxed-answer probe), and an *adaptive* bank that additionally fires event-triggered probes at entropy drops and at conclusion/reflection/answer markers.

3.2 Token accounting

For a rule that stops a problem at token position s after consuming probe output p, we charge total decode tokens $T = s + p$ and compare against the frozen baseline B (the full trajectory, capped at budget). *Net* savings is $(B - T)/B$ and *gross* savings is $(B - s)/B$; their gap is the probe cost. The accuracy drop compares the correctness of the committed answer at s against that of the frozen final answer (it is not an agreement rate between them).

3.3 Models, benchmarks, splits

Development uses two models—DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B—on three benchmarks (MATH500, AMC23, AIME24), each at seeds 42/43/44: 18 model×benchmark×seed environments. Problems are partitioned *by problem id* into 60/20/20 train/dev/test splits, so no problem leaks across splits. Confirmation reserves seeds 45/46/47 plus two unseen models—DeepSeek-R1-Distill-Llama-8B and Distill-Qwen-32B—all evaluated on the *test* split only, read once after all rules and thresholds are frozen, never for tuning. Every headline metric is a *macro*-average over model×benchmark×seed environments, weighting each equally, so that the largest benchmark does not dominate policy selection (Appendix B details the weighting and its noise trade-offs).

4 The Consensus–Termination Gap

This section establishes the paper's central object by direct measurement, before any rule sweep; the exhaustive search of Section 5 rests on what is measured here. The source of the gap is that a probe does not read a settled belief—it requires an answer even when the underlying trajectory has not settled, so repetition under a fixed query may reflect the elicitation procedure rather than reasoning termination: an elicited placeholder that can look stable while the trajectory keeps correcting itself. We establish, in order: agreement is pervasive and non-terminal (§4.1); the early answer depends on how the probe is worded (§4.2); most stopped-but-wrong commitments are to answers the model had not converged on (§4.3); the resulting cost is *directional* (§4.4); and non-independent readings supply no vote that could catch the error (§4.5).

4.1 Agreement Is Not Terminal

An online rule cannot wait for a trajectory to end: it fires the *first* time its condition is met, so what matters is what the *first actionable consensus* commits to. On a dense probe stream logged for every MATH500 problem under DeepSeek-R1-Distill-Qwen-7B at the main 16K/32K budgets (500 problems × three seeds = 1,500 trajectories, never intervening; descriptive only, and it does not touch the test commitment; Appendix G), we take that to be the first point at which three consecutive probes agree, are non-empty, and satisfy the certainty requirement (a simplified heuristic, not the CertaIndex (CoT) rule of §5.4).

(1) The first actionable consensus is usually not terminal. It fires on $1{,}477/1{,}500$ trajectories, and the answer it commits is correct only 50.5% of the time—an error rate of 49.5%—whereas the same trajectories run to completion reach 90.7%: a 40.2 pp loss.

(2) End-of-trajectory agreement is a different object. Once a trajectory finishes, agreement does track correctness (97.8% for cumulative agreement across all probes, 90.4% for the final five-probe window), but no such end-state evidence is available mid-trajectory. Recovery separates the two: among the 736 trajectories that *at some point* formed a three-probe consensus different from their eventual final answer, 84.2% ultimately ended with a correct *final* answer—a statement about where continued generation arrives, not about the accuracy of the answer a stop would have committed.

The same question on a broader environment set. Facts (1)–(2) read one model on one benchmark. We therefore repeat the measurement across all 18 development environments (both models, three benchmarks, three seeds, dev split; 684 trajectories), this time dropping the certainty requirement so that the rule fires as early as it can. First consensus forms in 679 of the 684 trajectories, half of them within the first tenth of the trajectory's own length. There the committed answer is correct only 27.5% of the time against 85.2% run to completion, and the gap narrows the later first consensus falls (Figure 4, appendix). Reliability therefore *improves* with relative position: the failure is not that late consensus is bad but that an online rule acts on the *earliest* consensus, which commonly appears before the trajectory has settled.

4.2 Probe Wording versus Position

If early agreement reflected a settled belief, the answer a probe returns should not depend much on how the probe is worded. We test this directly on paired re-probes of the *same frozen prefixes*: at every probe position we read the trajectory with two different probe wordings (our boxed-answer suffix and the CertaIndex suffix, both capped at 32 tokens), so the model's state is identical and only the elicitation differs. Scope: an exploratory pass on one environment (DeepSeek-R1-Distill-Qwen-7B × MATH500, seed 42) re-probed out to 3,072 tokens—a shorter budget than the 16K/32K main trajectories—read on the 241 of 400 trajectories it covers end to end (2,898 positions), which are therefore the *shorter* ones. This section and §4.3 are diagnostic: nothing in §5 rests on them.

Figure 2(a) shows the result. In the first tenth of a trajectory the two wordings return different answers 54% of the time; by the final third they disagree only 10% of the time. Early "answers" are therefore substantially a property of the question asked, and become properties of the state being read only as the trajectory finishes.

This rules out the obvious alternative: if the model held a stable but mistaken belief early on, both wordings would read back the *same* wrong value; instead they disagree. What a consensus rule treats as a converging answer is, early on, an output the probe has elicited. Probe wording has almost no effect on *final* accuracy in the same paired data (a 0.65 pp readout effect against a 9.15 pp effect of *when* one probes), so the sensitivity is specific to early positions, not a defect of one suffix.

4.3 Error Taxonomy

Two annotators independently labelled *all* 134 cases in which probes agreed on a wrong answer, from the same exploratory pass as §4.2 (DeepSeek-R1-Distill-Qwen-7B, MATH500, seed 42, 3,072-token cap); agreement on the top-level coding (substantive error / probe-format artifact / other) is $\kappa = 0.82$. Only 17.9% are cases where the model had genuinely settled on a wrong value (Figure 2(b)). 61.2% are answers it had not converged on at all, and a further 18.7% are artifacts of the probe's output format.

The reason is that a probe does not read the model's belief; it *forces* an answer. Because a probe requires an answer even when the underlying trajectory has not settled, repetition under a fixed query may reflect the elicitation procedure rather than reasoning termination, and a consensus rule cannot tell that case apart from a settled one. This also predicts the window effect of §4.4: a longer window is a longer buffer in which a placeholder can be overturned, which is why it screens out some non-terminal agreement without removing the underlying risk.

4.4 A Directional Cost, Not Sampling Noise

The natural remedy is to demand more agreement before stopping—a stop only when the last W probes agree, for larger W. It helps, but only by stopping less often. Consider only the stops where committing early *changes* correctness relative to the trajectory's own final answer, and count two outcomes: (final-correct, stop-wrong)—a correct recovery destroyed—versus (final-wrong, stop-correct)—a wrong answer luckily banked. Pure sampling noise predicts a $\approx 1{:}1$ ratio; a genuine "continued reasoning corrects" effect predicts $\gg 1$. On the frozen replay this ratio depends strongly on the window: it falls from $\approx 45{:}1$ for a latest-probe stop ($W{=}1$) to $\approx 2{:}1$ at the largest window ($W{=}30$). But the low end is bought only by stopping so rarely that net saving vanishes ($W{=}30$ fires on 121 dev problems vs. 668 at $W{=}1$); at any window that actually saves tokens the ratio stays well above 1:1. DEER (the non-consensus control of §5.3), by contrast, holds ≈ 2.4–$3.5{:}1$ at its three operating points *while* saving 28–32%, because its boundary-confidence signal commits mainly once the trajectory has settled. The cost of stopping on consensus is therefore directional rather than sampling noise—the recovery phenomenon of §4.1 as a signed effect. Figure 3 plots the whole window axis, so the claim rests on the shape of the curve rather than on either endpoint: the ratio and the saving fall together.

Why does the ratio stay above one wherever the rule still fires? For a stop at token position s, the correction forgone is whatever the trajectory would have made after s—a property of the trajectory, not of how it was probed. A rule's *realized* cost does depend on where it stops, and hence on the probe schedule, but the asymmetry producing that cost is present at every position where reasoning is still moving. Appendix F shows all three regimes on individual probe streams, including a first-probe placeholder held for 27 consecutive probes—long enough to outlast all but the largest window we searched.

4.5 The Role of Independence

A structural fact explains why self-consensus has no accuracy backstop: a wrong stop is never overturned, which is part of why the accuracy loss is so large. Self-consistency's majority vote can absorb an error because its paths are sampled independently (§2): a wrong answer on one path is outvoted by others that reasoned differently. Probing one trajectory yields no such thing—successive probes observe nested prefixes from the same evolving reasoning path and therefore share most of their reasoning history, so the readings move together,

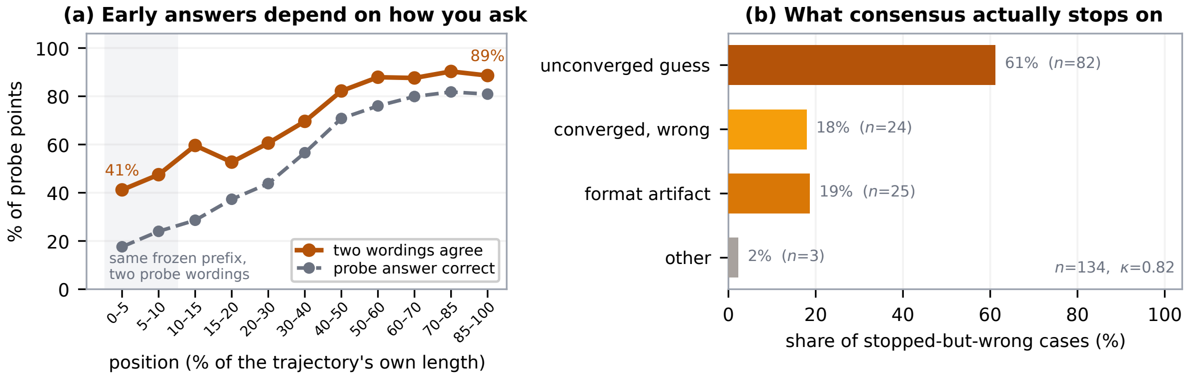

Figure 2: **Two sides of the consensus–termination gap. (a)** The same frozen prefix (DeepSeek-R1-Distill-Qwen-7B, MATH500) re-probed with two differently worded answer queries. Early on the two wordings return the *same* answer less than half the time (41% in the earliest bin, 46% over the first tenth); by the end they nearly always coincide (90%). Position is a fraction of each trajectory's own length. **(b)** All 134 stopped-but-wrong cases, labelled independently by two annotators (κ=0.82 on the top-level coding): 61.2% commit to an answer the model had *not converged* on, 18.7% to a probe-format artifact, 17.9% to a genuinely settled wrong value.

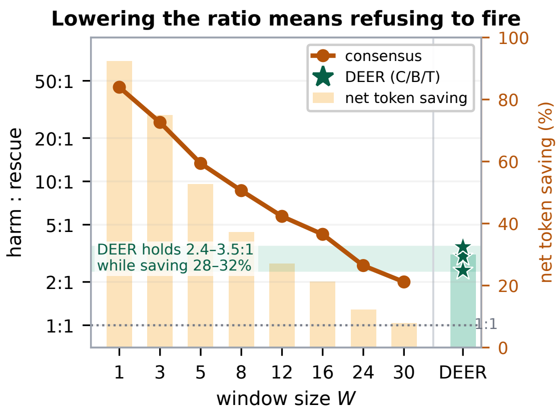

Figure 3: **Lowering the ratio means refusing to fire.** Recoveries destroyed vs. wrong answers banked, among stops whose committed answer differs in correctness from the trajectory's final answer (sampling noise predicts 1:1). Widening the window brings the ratio from 45:1 to 2:1 only by stopping on ever fewer problems ($668 \to 121$ of 684); net saving collapses from 92% to 8% alongside. Dev split, 18 environments; one rule per W (share 1.0, interval 128, maturity 512, schema validity).

and when that shared history is wrong they are wrong together. Agreement across probes is thus a statement about one line of reasoning not having changed its answer, not corroboration from several: a consensus rule holds no vote in reserve to overturn the placeholder it commits to. This explains why the gap is *dangerous*, not that it exists—that is what the measurements above establish.

Whether some setting of the consensus signal nonetheless escapes the gap is an empirical question; the next section closes it.

5 The Gap Cannot Be Tuned Away

Section 4 measures the gap. The natural response is that a better-tuned rule avoids it—a wider window, a stricter share, a maturity floor, a certainty requirement. This section rules that out by exhaustive search under commitment: a *preregistered* sweep of the consensus family with acceptance gates fixed in advance (Nosek et al., 2025), a non-consensus signal swept through the same pipeline, a released consensus stopper at its shipped default, and out-of-distribution confirmation. Two questions run through it. First, whether *any* rule in the family is simultaneously safe and saving on the split the gates are applied to. Second, if one were, whether it would be an *environment-robust policy*—selected once rather than tuned per environment, and still safe and saving on held-out problems, seeds, benchmarks and models.

5.1 Preregistered Rule Space and Gates

The consensus signal in the windowed family we study reduces to two hyperparameters: a **window size** W (how many of the most recent probes to inspect) and a **share threshold** s (what fraction of that window must agree on a single answer before a stop is allowed). W=1 is the latest-probe rule; s=1.0 is strict unanimity over the last W probes (the CertaIndex-style stop); an entropy-over-the-window criterion closely tracks the same agreement and adds no materially new operating point on our grid. We sweep $W \in \{1, 3, 5, 8, 12, 16, 24, 30\}$—

Operating point	Max total acc. drop	Min total saving	psf floor
conservative	1.0 pp	10%	0.80
balanced	2.0 pp	20%	0.80
token_efficient	3.5 pp	30%	0.70

Table 1: Preregistered acceptance gates, fixed in advance and *not* relaxed post hoc. All three conditions are macro-averaged over the 18 development environments.

deliberately including *large* windows, so a rule can demand a long, stable agreement—and $s \in \{0.6, 0.8, 1.0\}$, crossed with the operational knobs that govern *when* and *how* the signal is read: probe schedule (four fixed intervals or event-triggered probing), a minimum-token maturity floor, an answer-shape validity filter, and an optional certainty flag. Enumerating this preregistered grid yields 3,520 candidate consensus rules (all values and formal definitions in Appendix A).

A non-consensus control. To separate a failure of the consensus family from a failure of early exit itself, we sweep one non-consensus method through the same pipeline, gates, and token accounting. **DEER** (Yang et al., 2025) stops on the model's own confidence in a trial answer at a reasoning boundary; its only hyperparameter is a confidence threshold, swept over 14 values (the stronger *trial-answer-submit* variant). Both are placed on one frontier under identical token accounting. It is a control on the family, not a one-factor ablation of the statistic (§5.6).

Gates. We fix three operating points and their acceptance gates *before* looking at held-out data, applied in order: the total accuracy drop (macro over the 18 environments) must stay under a cap, then the total net token saving must clear a floor, and finally net saving must be positive on a minimum fraction of environments (psf).

The three points (Table 1) pair progressively looser accuracy constraints with progressively higher saving requirements. They are not universal safety standards; they test whether self-consensus supplies a usable policy under *some* practically meaningful pairing of the two. The saving floor rejects a rule that stays accurate only by almost never stopping, and the psf floor an average saving concentrated in a few environments. Three commitments were recorded with the protocol: the test split and confirmation models are never read during selection (C1); the gates are never relaxed post hoc—an empty gate is reported as the finding (C2); and the rule set is frozen and hashed before any test evaluation (C3). Appendix B gives the preregistration text.

5.2 No Rule Is Both Safe and Saving

We replay all 3,520 rules on the frozen trajectories and aggregate to per-environment metrics—126,720 rows = 3,520 rules $\times$ 18 environments $\times$ 2 splits (train, dev)—then apply the gates on dev. **Not one of the** 3,520 **rules clears any gate.** The outcome holds across environments: accuracy strictly falls in 64.9% of all rule–environment evaluations and strictly rises in only 10.5%, for a mean drop of 13.0 pp. The reason is the shape of the trade-off, read directly off the sweep's own two axes (Figure 5, appendix): only 40 rules keep the total drop at or below the conservative 1.0 pp cap, and the most any of them saves is 0.2%—far under the 10% floor; conversely, the first rule to save 10% already costs 2.66 pp, saving 20% costs 6.17 pp, and saving 30% costs 11.8 pp. The large windows behave exactly as §4.4 predicts: they do buy low drop, but only by stopping so late that net saving collapses to zero, and no cell of the (W, s) surface clears the conservative gate. Nor is this an artifact of charging dense re-probing: that *probe tax* opens only a 4–7 pp gap between gross and net saving and is reducible by a sparser schedule, whereas the accuracy floor the gates bind on is not (Appendix D).

5.3 A Non-Consensus Signal Clears the Same Gates

The same gates are cleared, comfortably, by a rule that does *not* read consensus. Sweeping DEER's confidence threshold through the same pipeline and accounting, its trial-answer-submit variant passes all three operating points on dev: the conservative point loses only 0.33 pp while saving 28.2%, the balanced point 1.03 pp at 29.6%, and the token-efficient point 2.75 pp at 31.9%; at a stricter threshold it is accuracy-neutral (+0.06 pp) while still saving 20.8% (Table 2; full threshold frontier in Appendix D). Early exit is therefore not impossible at these budgets; the consensus *signal* is what cannot make it safe.

5.4 Consensus in the Wild

The sweep covers rules we constructed; a released consensus stopper shows where a shipped default lands. **CertaIndex (CoT)** (Fu et al., 2024), reproduced from the released implementation at its

Method	Acc.	Δacc (macro)	Net saving
Full generation	82.5%	—	0%
Consensus (drop≤ 1.0)	81.6%	-0.9 pp	$+0.2\%$
Consensus (save$\geq 10\%$)	79.9%	-2.7 pp	$+10.9\%$
DEER (conservative)	82.5%	-0.33 pp	$+28.2\%$
DEER (balanced)	81.9%	-1.03 pp	$+29.6\%$
DEER (token_eff.)	80.1%	-2.75 pp	$+31.9\%$

Table 2: Dev operating points (macro over 18 environments). No consensus rule is simultaneously safe (≤ 1.0 pp) and saving ($\geq 10\%$): capping the drop forces saving to near zero, and demanding saving forces a large drop. DEER, reading a non-consensus signal, clears all three gates on the same trajectories under the same accounting.

default setting—stop once three consecutive probes give the same non-empty answer and none hedges, with the authors' own probe prompt and schedule—stops on 98.7–99.8% of our frozen trajectories and loses 56–70 pp while saving 77–90% of tokens. This is where the swept frontier says a rule at that stop rate must land: the trade-off of §5.2 at one shipped operating point, not an implementation defect. A faithful DEER *readout* reproduction on the same trajectories stays near full accuracy ($+0.8$ pp on Qwen3-8B). Table 4 (appendix) reports both reproductions with scope notes; in particular, nothing here bears on CertaIndex's multi-path setting, which is ordinary self-consistency over independently sampled trajectories.

5.5 Generalization

We read the test split once (commitment C3); with no selected rule to confirm, we ask whether the frontier *reproduces*. It does. Each rule's total drop on the held-out test split tracks its dev value at $r{=}0.98$, and the joint conservative gate stays empty: 0 rules clear it on both splits, while 444 rules are admissible on test, 364 of them the in-sample winners a held-out split exists to reject. On the two unseen models—Distill-Qwen-32B (4× the development scale) and DeepSeek-R1-Distill-Llama-8B (a different architecture and family), three test seeds each—the frontier reproduces at $r{=}0.97$ and $r{=}0.94$, and the conservative gate stays empty on every model. DEER, swept identically, clears the gates everywhere: on dev and test, and on both unseen models (accuracy-neutral at -0.24 pp while saving 32.4% on the 32B; 0.67 pp at 26.7% on Llama)—across every axis we vary: benchmark, seed, 4× scale, and architecture. Appendix E traces the selection pipeline end to end (Figure 6), gives per-model and per-benchmark frontiers with an oracle upper bound, and details the dev/test asymmetry induced by the small competition sets.

5.6 Locating the Failure

Every method that stops on *consensus*—the swept family and the CertaIndex (CoT) reproduction alike—fails a gate or loses many points, while the one method that clears them does not read consensus at all. Run through the same machinery, gates, and accounting, the two outcomes differ, so the failure lies in how the stop is decided rather than in early exit at these budgets.

Four qualifications bound the claim. It concerns agreement used *alone*: combined with a signal reflecting the model's own estimate, the cheap fact that the answer has not changed may well be useful, as CertaIndex's composite indicator assumes (Fu et al., 2024). It concerns the preregistered schema and dense-probe regime we searched—our probe grid makes essentially every stop position reachable, so the floor is not a lucky miss of the schedule, but a finite search demonstrates a systematic risk, not a universal impossibility theorem. The DEER contrast is not a one-factor ablation: DEER also differs in when it reads (reasoning boundaries, not a fixed schedule) and what it commits (a trial answer, not a probe answer), so it locates the failure in the consensus family *as deployed*, not in the agreement statistic in isolation. And it does not say which signal *should* replace agreement; that, with its own train/dev→test confirmation, is left open.

6 Conclusion

Used on its own, self-consensus is not a safe early-exit signal: under a fixed probing procedure agreement establishes that the current answer *persists*, not that the reasoning has *terminated*—a *consensus–termination gap*. What the probes agree on is often a placeholder forced from an unfinished trajectory, so stopping destroys far more correct-in-the-end answers than it rescues wrong ones, and more agreement only trades stopping opportunities away. That asymmetry leaves the safe-and-saving corner empty across 3,520 preregistered rules, while a boundary-confidence control swept through the same protocol clears every gate. Agreement fails not because it is insufficiently strict, but because it repeatedly measures the wrong object.

Limitations

The mechanism is argued, not isolated experimentally. Our account of *why* consensus fails—that probes of one trajectory are not independent readings, and that they frequently record a forced placeholder—rests on a structural argument (Sections 2 and 4.5) together with the hand-labelled error taxonomy and the window sweep. We do not run a controlled experiment that manipulates independence directly, and the labelled cases were collected under a short window, the regime where unconverged answers are most common. A stronger test would construct probe streams with varying degrees of dependence and measure how the accuracy tax responds.

Scope of the negative result. The central result—no consensus rule clears any gate—is established on 18 development environments (two models, three benchmarks, seeds 42/43/44) and confirmed on the held-out test split and two unseen models (Section 5). It is a statement about the space searched: one boxed-answer probe suffix (simple@32) and one preregistered consensus schema (3,520 rules over a window-size $\times$ share-threshold family plus operational knobs). Other probe wordings and signals outside this schema are not searched; the mechanism (Section 4) is our argument that consensus variants would not help, but it is an argument, not an exhaustive proof.

Scope of the DEER comparison. We sweep DEER's trial-answer-submit variant to establish that *some* signal clears the gates the consensus family fails (on the development models, the held-out test split, and both unseen models); we do not claim it is the best possible early-exit method, and we do not evaluate it as a leaderboard entry.

Small held-out sets on hard benchmarks. AIME24 and AMC23 are small (dev splits of 6 and 8 problems per seed), so those per-environment cells are noisy. The total drop macro-averages over the 18 development environments (three seeds per benchmark$\times$model), and we read the unseen models as frontier-reproduction evidence rather than as independent gate tests (Section 3.3).

Probe-tax dependence of the savings axis. Net-savings numbers charge dense re-probing (every 64 tokens, 32-token probes); a sparser or KV-cache-reusing schedule would change the *savings* axis and could move some rules to positive net savings. We separate gross from net savings (Section 3.2) precisely so the probe-tax-dependent part is distinguishable; the accuracy-tax result does not depend on the schedule.

Domain. All benchmarks are competition mathematics with checkable final answers. Whether the consensus–termination gap and the accuracy tax transfer to open-ended reasoning, code, or agentic tasks is out of scope.

References

Pranjal Aggarwal, Aman Madaan, Yiming Yang, and Mausam. 2023. Let's sample step by step: Adaptive-consistency for efficient reasoning and coding with LLMs. In *Proceedings of EMNLP*.

Xingyu Chen, Jiahao Xu, Tian Liang, Zhiwei He, Jianhui Pang, Dian Yu, Linfeng Song, Qiuzhi Liu, Mengfei Zhou, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. 2024. Do not think that much for 2+3=? on the overthinking of o1-like LLMs. *arXiv preprint arXiv:2412.21187*.

DeepSeek-AI. 2025. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*.

Zhe Dong, Fang Qin, and Manish Shah. 2026. When does learning to stop help? a cost-aware study of early exits in reasoning models. *arXiv preprint arXiv:2606.30852*.

Yichao Fu, Junda Chen, Siqi Zhu, Zheyu Fu, Zhongdongming Dai, Aurick Qiao, and Hao Zhang. 2024. Efficiently serving LLM reasoning programs with Certaindex. *arXiv preprint arXiv:2412.20993*.

Alex Graves. 2016. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *NeurIPS Datasets and Benchmarks Track*.

Parsa Hosseini, Sumit Nawathe, Mahdi Salmani, Meisam Razaviyayn, and Soheil Feizi. 2026. Early stopping for large reasoning models via confidence dynamics. *arXiv preprint arXiv:2604.04930*.

Yiwei Li, Peiwen Yuan, Shaoxiong Feng, Boyuan Pan, Xinglin Wang, Bin Sun, Heda Wang, and Kan Li. 2024. Escape sky-high cost: Early-stopping self-consistency for multi-step reasoning. In *International Conference on Learning Representations (ICLR)*.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John

Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*.

Xin Liu and Lu Wang. 2025. Answer convergence as a signal for early stopping in reasoning. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Brian A. Nosek et al. 2025. Preregistration in machine learning research. *arXiv preprint*.

OpenAI. 2024. Learning to reason with LLMs. https://openai.com/index/learning-to-reason-with-llms/.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. 2022. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. 2024. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *Proceedings of ACL*.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*.

Chenxu Yang, Qingyi Si, Yongjie Duan, Zheliang Zhu, Chenyu Zhu, Zheng Lin, Li Cao, and Weiping Wang. 2025. Dynamic early exit in reasoning models. *arXiv preprint arXiv:2504.15895*.

A Rule Schema Details

The consensus schema of Section 5.1 centers on two hyperparameters and a few operational knobs, enumerating to 3,520 candidate rules over a fixed-schedule and an event-triggered family. Each rule fixes:

1. **Window size** $W \in \{1, 3, 5, 8, 12, 16, 24, 30\}$: the number of most recent probes inspected. $W{=}1$ is the latest-probe rule.
2. **Share threshold** $s \in \{0.6, 0.8, 1.0\}$: the fraction of the last W probes that must agree on one answer (share measured over the full window, so an empty or dissenting probe counts against consensus); $s{=}1.0$ is strict unanimity. An entropy-over-the-window criterion closely tracks the same agreement and adds no materially new operating point on our grid, so it is not swept separately.
3. **Probe schedule**: fixed interval $\in \{64, 128, 256, 512\}$ tokens, or `adaptive_event` with entropy-drop and marker triggers and a fallback interval.
4. **Maturity**: a minimum-token floor $\in \{0, 512, 1024, 2048, 4096\}$ before any stop is allowed.
5. **Certainty**: optionally require the supporting probes flagged certain (minimum fraction).
6. **Validity**: answer-shape filter (reject empty / single-letter answers to suppress the Type-E probe artifact of Appendix G), or accept any non-empty answer.

Certainty. A probe is *certain* when its boxed answer is emitted with high token-level confidence: concretely, the mean probability of the answer tokens under the probe forward pass exceeds a fixed threshold (the Certaindex certainty flag). For the simple@32 bank this is aggregated over the 32 samples as the fraction agreeing on the modal answer. Dimension 5 gates a stop on the fraction of the supporting probes that are certain.

Entropy triggers. The `adaptive_event` schedule fires an extra probe when the token-level predictive entropy of the main decode drops by at least ΔH relative to a trailing baseline (an "the model just committed" signal), and at conclusion/reflection/answer marker tokens (e.g., "therefore", "the answer is"), subject to a minimum inter-probe gap and a fallback interval.

B Preregistration Text

The protocol version, split manifest hash, candidate-rule generation script, and the three operating-point gates (Table 1) were fixed before any held-out evaluation. The commitments of Section 5.1 in full: **(C1)** the test split and both confirmation models are never read during rule selection. **(C2)** All three operating points and their gates are fixed before seeing held-out data and are not relaxed afterwards; if no rule passes a gate, that is reported as the finding rather than engineered away, and all three points are reported regardless of outcome. **(C3)** The frozen rule set is fixed on train+dev and its hash recorded before any test/confirmation evaluation. When the conservative gate proved empty we followed C2 and report the empty gate as the finding rather than relaxing it, for both consensus and DEER.

Metric resolution. The total accuracy drop is a macro-average over the 18 environments (each model×benchmark×seed weighted equally). Pooling problems instead would let MATH500 (100 dev problems per seed vs. 8 for AMC23 and 6 for AIME24) dominate the headline, so a rule that wins only on MATH500 could pass; equal per-environment weight forces a selected rule to hold on the harder competition sets too; three seeds per cell damp the per-environment noise. The gates are read on the dev split (the selection target) with train reported alongside; where the text says "train+dev," a rule was required to pass a gate in-sample on train and is then measured on dev. Because *no* consensus rule clears any gate, by a wide margin, the finding does not rest on a single fragile cell.

C Reproducibility

Frozen trajectories, offline probe banks, the sweep archive (126,720 dev metric rows plus the DEER threshold sweep), and the aggregation scripts are released. Table 5 and Figure 5 were reproduced to the reported precision directly from the released sweep archive. Grading uses a robust answer-equivalence check that tries the raw and normalized reference forms and both string and math-expression equality; per-problem correctness for the frozen baseline is recomputed at replay time rather than trusted from collection.

D Frontier and Reproduction Tables

Figure 5 traces the consensus trade-off along both sweep axes; read down the window axis, the minimum attainable accuracy drop falls monotonically (from 22 pp at W=1 to 0.8 pp at W=30) while the maximum net saving falls with it (from 98% to 39%), so the low-drop region is reachable only where saving has already collapsed. Table 5 gives the DEER threshold frontier that does reach the gate region.

Accuracy tax versus probe tax. Table 3 separates the two costs of §5.2: under dense re-probing a late-stopping rule pays the probe output p of Section 3.2 over a long prefix, opening a gap between its gross savings $(B - s)/B$ and its net savings $(B - s - p)/B$. That gap—the probe tax—scales with probe density and is reducible: sparser or shorter probes and KV-cache reuse all shift the *savings* axis. The *accuracy* floor is not reducible this way—the recovery asymmetry behind it is present wherever reasoning is still moving (§4.4)—and it

Consensus point	Gross saving $(B - s)/B$	Net saving $(B - s - p)/B$
save≥ 10% (drop 2.7 pp)	+14.9%	+10.9%
save≥ 20% (drop 6.2 pp)	+27.0%	+20.2%

Table 3: Gross vs. net savings for two consensus frontier points; the gap ($\sim$ 4–7 pp) is the probe tax. It shifts the *savings* axis but never rescues the accuracy drop—which is the intrinsic tax that keeps the corner empty.

is that floor, not the probe tax, that keeps the safe-and-saving corner empty and that DEER escapes.

Prior-stopper reproductions. Table 4 reports the two reproductions of §5.4 in a common harness—the same frozen trajectories, the same token accounting. **CertaIndex (CoT)** (Fu et al., 2024) is the consensus stop for a single chain of thought, reproduced from the released implementation at its default `mid` setting (stop once three consecutive probes give the same non-empty answer and none hedges), with the authors' probe prompt and probe schedule—their rule as shipped, not a variant of ours. **DEER** (Yang et al., 2025) here is the faithful readout reproduction (the stronger trial-answer-submit variant is what §5.3 sweeps). Two scope notes. First, CertaIndex's multi-path setting is ordinary self-consistency: it measures entropy over answers from *independently sampled* trajectories, and nothing here bears on it; what we reproduce is the single-trajectory case, where the same statistic is read over repeated probes of one chain. Second, these are reproductions on our frozen trajectories and benchmarks, not re-runs of either paper's end-to-end system, and the original CoT results were reported on an easier benchmark, where intermediate answers settle sooner.

E Supporting Figures: Sweep Surface, Selection, and Frontiers

This appendix collects the supporting figures referenced from the main text. Figure 4 shows where consensus first forms as a fraction of each trajectory's length (§4.1); Figure 5 reads the empty safe-and-saving corner directly off the sweep's (W, s) surface (§5.2); and Figure 6 traces the train → dev → test selection pipeline (§5.5). Figures 7–9 then give the saving-versus-drop view of the same selection and generalization results, resolved per split, per model and per benchmark, with the oracle upper bound on each panel. They add no claim beyond the main text; they show the

Method	Model	Accuracy	Δacc	Fair token saving	Stop rate	Signal
Full generation	Qwen3-8B	85.4%	—	0%	0%	—
Full generation	DeepSeek-R1-Distill-Qwen-7B	79.8%	—	0%	0%	—
CertaIndex (CoT)	Qwen3-8B	15.3%	−70.1 pp	90.1%	99.8%	consensus
CertaIndex (CoT)	DeepSeek-R1-Distill-Qwen-7B	23.9%	−55.9 pp	76.7%	98.7%	consensus
DEER	Qwen3-8B	86.2%	**+0.8** pp	16.3%	41.4%	boundary conf.
DEER	DeepSeek-R1-Distill-Qwen-7B	74.9%	−4.8 pp	20.2%	56.1%	boundary conf.

Table 4: Prior early-stoppers reproduced on our frozen trajectories (dev, macro over three benchmarks; probe tokens charged). **CertaIndex (CoT)** stops on almost every problem and stops early, and loses 56–70 pp while saving 77–90% of tokens—the pattern §4.1 describes, at a single shipped operating point. **DEER**, reading boundary confidence rather than consensus, stays near full accuracy (+0.8 pp on Qwen3-8B) while still saving tokens. These are reproductions on our benchmarks, not the original papers' end-to-end numbers.

Threshold τ	Total drop (pp)	Net saving
0.9999	−0.06	20.8%
0.999	0.06	25.4%
0.995	0.33	28.2%
0.99	1.03	29.6%
0.97	2.75	31.9%
0.95	5.87	34.8%
0.90	11.43	42.6%

Table 5: DEER trial-answer-submit threshold frontier on dev (macro over 18 environments, same token accounting). Unlike consensus, low drop coincides with substantial saving—the conservative/balanced/token-efficient gates are all cleared around $\tau \in [0.97, 0.995]$.

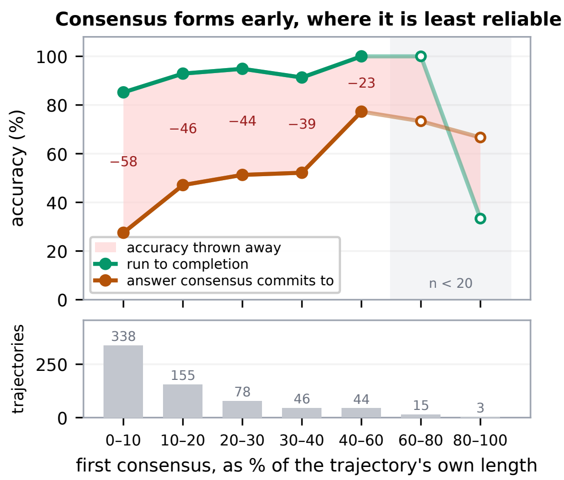

Figure 4: **Consensus forms early, where it is least reliable.** For each dev-split trajectory we take the first window of three agreeing non-empty probes and ask two things: is that answer correct, and would the same problem have been correct if left to run? Position is a fraction of the trajectory's *own* length, so 500 tokens into a 600-token trajectory counts as late and 500 into a 16K one does not. Consensus forms in 679 of 684 trajectories, half within the first tenth; the shaded gap is the accuracy a stop at that moment throws away. Open markers mark bins with fewer than 20 trajectories; the final bin (n=3) should not be read.

distributions the summaries are drawn from.

Selection-pipeline details (§5.5). In Figure 6, the 478 consensus rules that clear the conservative gate *in-sample on train* are tracked onto dev and test as a band; the three dev-selected DEER thresholds are tracked individually. On dev—the split the gate is applied on—not one of the 478 is admissible, while all three DEER points stay inside on every split. The dev and test splits are not equally demanding: the same 478 rules have a median drop of 4.5 pp on dev but 0.6 pp on test, an asymmetry driven by the two smallest environments (AMC23 and AIME24 under Qwen3-8B). This is why the claim is stated jointly rather than per split: 364 of the 478 are admissible on test, but they are exactly the in-sample winners a held-out split exists to reject, and none of them is selectable from dev.

Unseen-model details. On the 4× larger Distill-Qwen-32B the conservative gate stays empty (the best rule under a 1.0 pp drop saves 0.6%), but a scale effect appears at the looser gates: a few consensus rules become admissible *in-sample* (4 at balanced, 6 at token-efficient), reflecting that a bigger model's intermediate answers stabilize somewhat earlier—these are not the dev-selected rules (dev admits none). On the same-scale Distill-Llama-8B all three gates stay empty (0/0/0; the best rule under a 1.0 pp drop saves 9.3%). The Llama-8B trajectories were re-collected after correcting a chat-template defect (a missing model-specific beginning-of-sequence token that degraded its generations); the corrected model scores ∼ 88% on MATH500. The *oracle* upper bound on each panel is a perfect probe-based stop committing the earliest correct probe; it sits at

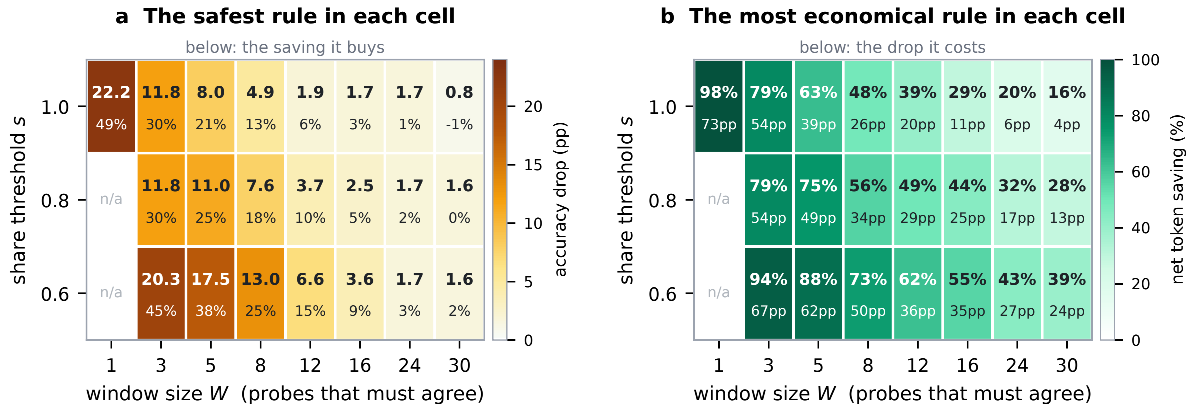

Figure 5: **The safe-and-saving corner is empty across the entire rule space.** Dev split, macro-averaged over the 18 environments. Each cell holds the 160 rules sharing its (W, s), summarised by the two members a gate interrogates: **(a)** the cell's *safest* rule (colour: accuracy drop; number: the saving it buys) and **(b)** its *most economical* rule (colour: net saving; number: the drop it costs). A cell would be outlined in green if any of its rules cleared the conservative gate (drop ≤ 1.0 pp *and* saving $\geq 10\%$); none is.

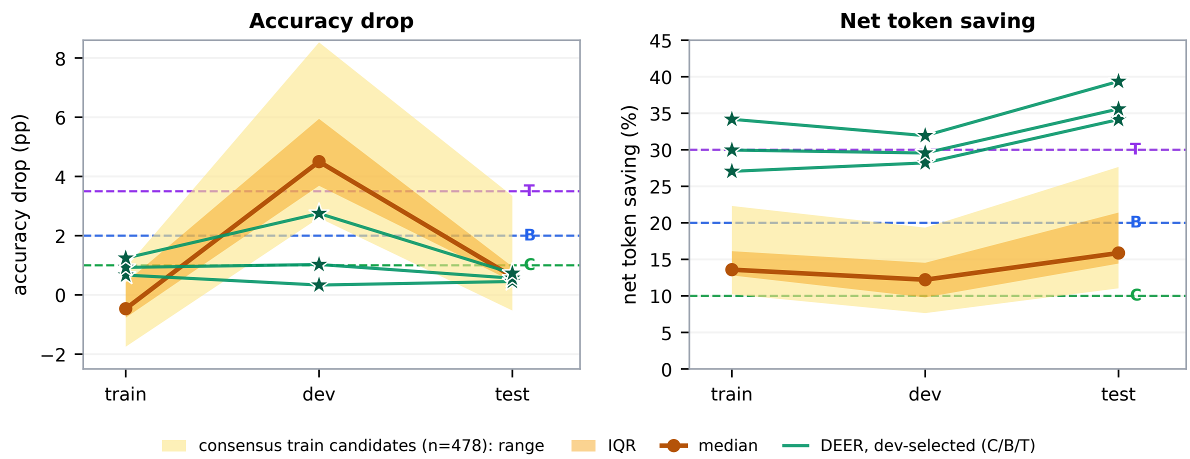

Figure 6: **A rule must clear the gate on every split.** The 478 amber rules are those clearing the conservative gate *in-sample on train*, shown as median, interquartile range and full range. Dashed lines mark the three preregistered gates (**C**onservative, **B**alanced, **T**oken-efficient); a rule must sit below them on the left and above them on the right. On dev—the split the gate is applied on—none of the 478 does; jointly, 0 of 478 clear all three splits, against 3 of 3 for the dev-selected DEER points. Macro-averaged over 18 environments per split.

a negative drop of 2–5 pp at 40–80% saving, far in a corner neither method reaches.

F Case Studies

The mechanism of Section 4 is easiest to see on individual trajectories. All three below are DeepSeek-R1-Distill-Qwen-7B on MATH500, seed 42, probed every 64 generated tokens; we write $a \times n$ for an answer repeated over n consecutive probes. A rule requiring three agreeing probes fires at the third element of the first such run.

Case 1: a placeholder that outlasts every window we searched. Problem 320 asks for $\cos \angle RPS$ given $\sin \angle RPQ = \frac{7}{25}$ (answer $-\frac{24}{25}$). The probe stream is

```
0×27, B×8, -24/25, B×7,
E, B×6, -24/25×7
```

The model answers 0 at the very first probe and repeats it for 27 consecutive probes, covering tokens 64–1728 of a 3,683-token trajectory. It then passes through a stretch of bare option letters before reaching $-\frac{24}{25}$ and holding it to the end; the trajectory is correct.

This single case makes three points. The stop is not a considered wrong answer but a value emitted before any real work has been done, and it is perfectly stable for nearly half the trajectory. That it

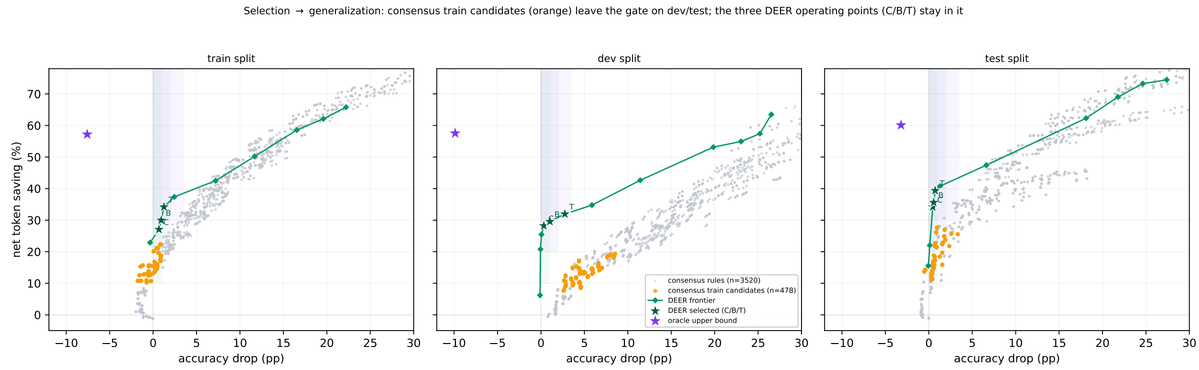

Figure 7: Selection → generalization across splits (two development models; macro-averaged over environments). Grey: all 3,520 consensus rules. **Orange**: the 478 rules that clear the conservative gate *in-sample on train*. **Green**: the DEER threshold frontier, with the three dev-selected operating points (**C**onservative/**B**alanced/**T**oken-efficient) as stars. **Purple star**: the oracle upper bound (earliest correct probe).

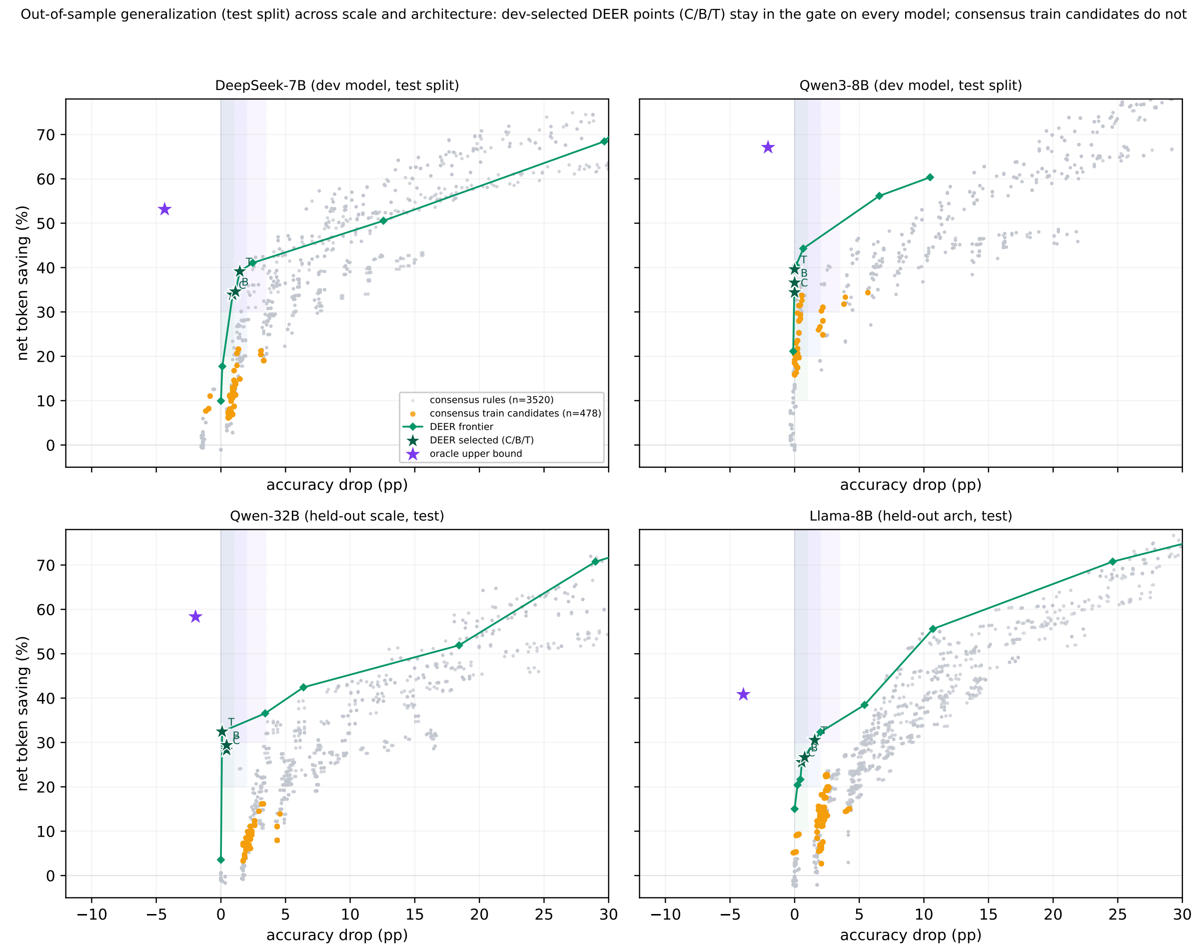

Figure 8: Out-of-sample generalization across scale and architecture (all panels test split). The two development models (top) are shown on their held-out test seeds; the two held-out models (bottom) are entirely unseen. Markers as in Figure 7. The dev-selected DEER points stay inside the conservative gate on every model; no consensus rule does.

never changes therefore says nothing about whether reasoning has finished. And because the run is 27 probes long, it defeats *every* window in our sweep: even $W{=}24$, the second-largest we searched, fires here and commits `0`. Enlarging the window does not address a placeholder that persists longer than

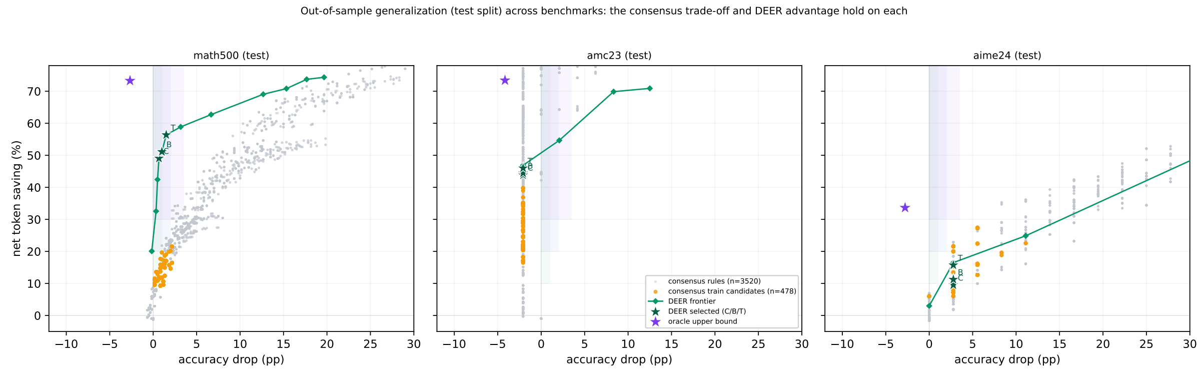

Figure 9: Out-of-sample generalization across benchmarks (test split, development models). The consensus trade-off and the DEER advantage hold on MATH500 and AMC23; on the small AIME24 set (few problems per environment) the two frontiers are noisier and intertwine. Markers as in Figure 7.

the window.

Case 2: the probe's format, not the model's answer. Problem 253 is "What fraction of 2 feet is 3 inches?"—a free-response question with no answer options. The stream is

`3, B, 3, 24, B, D×20, 1/8`

From token 384 to token 1600, more than two thirds of the trajectory, every probe returns the letter `D`. There is nothing for `D` to refer to. The model is not reporting a belief about the answer; it is completing the probe suffix in a plausible format while its reasoning continues underneath. The correct value $\frac{1}{8}$ appears only at the final probe. Agreement here is maximal and perfectly uninformative, which is why any deployed rule needs to check answer *shape* before treating agreement as evidence (§4.3).

Case 3: when there is nothing to recover. The minority case is equally worth stating. Problem 240 (gold 116) gives

`52×2, 104, 52×5, ..., 154×8, 52, 154×62`

A three-probe rule fires at token 384 and commits `52`; the trajectory itself ends at `154`. Both are wrong. Here early stopping destroys nothing, because continued reasoning was not going to arrive at the right answer either—the model settles firmly on `154` and holds it for its final 62 probes. Cases like this are why the harm-to-rescue ratio of §4.4 counts only stops where committing *changes* correctness: on problems the model was never going to solve, a consensus stop costs accuracy nothing and saves tokens, which is the regime consensus early exit was designed for. The difficulty is that agreement does not distinguish this case from Case 1, and Case 1 is the more common one.

G Self-Consensus: Full Analysis

Section 4.1 summarizes what this appendix establishes in full: in an intervention-free setting, self-consensus is not a reliable stopping signal.

G.1 Setup

We analyze the frozen main trajectories (Section 3): DeepSeek-R1-Distill-Qwen-7B on all 500 MATH500 problems (Lightman et al., 2024; Hendrycks et al., 2021) at the main 16K-token budget, temperature 0.6, top-p 0.95, across three seeds (1,500 trajectories in total).[1] The dense probe bank appends the boxed-answer probe suffix every 64 generated tokens and reads off the model's current answer. The probe is a separate forward pass: the controller only *observes*, never halting or altering the main trajectory. This yields, per trajectory, the full answer sequence a stopping rule would have access to at the budget used throughout the paper.

G.2 Final agreement is reliable; early agreement is not

Agreement measured at the *end* of a trajectory does track correctness. On the 186 trajectories that reach unanimous *cumulative* agreement (all non-empty probes concur), 97.8% are correct—only four whole-trajectory false consensuses; and of the 1,205 whose *final five-probe window* is unanimous, 90.4% are correct. But this end-state agreement is not what an online stopping rule can act on: a rule fires the *first* time it sees agreement, mid-trajectory, long before the answer has settled. The rest of this

[1] Train/dev problem ids are collected at seeds 42/43/44 and the held-out test ids at seeds 45/46/47; their union covers all 500 problems. This is a descriptive characterization of the frozen trajectories—it selects and tunes nothing—so it does not touch the preregistered sweep or the test-split commitment.

appendix shows that *early* agreement—the signal a controller must actually use—is far from terminal.

G.3 The cost lands at the stopping decision

We simulate a *naive consecutive-agreement* stop: halt at the first point where three consecutive probes agree, are non-empty, and are *certain* (the model emits the boxed answer with high token-level confidence; formal definition in Appendix A). This is a simplified heuristic, *not* the CertaIndex (CoT) rule, which we reproduce from its released implementation in §5.4. It fires on $1{,}477/1{,}500$ trajectories. The committed answers are correct only 50.5% of the time, whereas letting those same problems run to completion reaches 90.7%—a 40.2 **pp accuracy loss** in exchange for an average saving of 3301 tokens, with 731 trajectories stopped on an outright wrong answer.

G.4 Recovery is common, and early stopping kills it

The loss is driven by *recovery*: continued reasoning overturns the intermediate answer far more often than intuition suggests. Of the 736 trajectories that at some point formed a three-probe consensus *different* from their eventual final answer, 620 (84.2%) ultimately ended with a correct *final* answer—a statement about where continued generation arrives, not about the accuracy of the answer a stop would have committed. Local consensus is not close to terminal.

Normalizing by position *within each trajectory* shows where the damage concentrates: reliability *improves* the later the first consensus falls in a trajectory's own span, so the earliest consensus—the one an online rule acts on—is the least reliable (§4.1, Figure 4).